A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 35

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 B:** Oxygen	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) **1** There is still a lot that is not understood about how atmospheric processes work **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The surface warmed in the tropics heats the air above – warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes **1** Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** A fluid in a buried ground loop absorbs low-temperature heat from the ground – at the expansion valve the fluid boils at low pressure absorbing this heat **1** A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building **1** The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures **1** The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C **1** The stored heat boils water to drive steam turbines whenever electricity is required, including at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight – cloud, dust and smoke scatter light so it cannot be concentrated **1** PV cells convert diffuse as well as direct light, though output is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **diffuse** → pollution from a wide area with no single outlet	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1 Manufacture:** extraction / processing of silicon, metals and plastics – toxic wastes (silicon tetrachloride, cadmium) **1** Land take by large solar farms competing with other uses / affecting desert flora and fauna **1** Water for cleaning panels in arid areas **1** Disposal of old panels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change **1** Random sampling might leave gaps / miss the pattern – systematic sampling avoids bias while covering the gradient **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **systematic sampling** → sampling at regular intervals • **transect** → a line along which samples are taken to show change with distance • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data – wider intervals (eg 4 m) may miss peaks and troughs **1** Take increasing numbers of samples at one site and calculate the running mean – the number needed is where the mean stops fluctuating / stabilises **1** Calculate standard deviation to assess scatter: more variable data need more samples **1** Test the method (extraction technique, timing) for reliability before the main study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
03.3	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Remove surface vegetation – pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates **up to 2** Collect worms that come to the surface for a standardised time – count / identify / weigh **up to 2 Limitations:** solution may not percolate impermeable soils or reach deep worms – worms may move sideways or downwards – some may die or not respond – slow-moving worms missed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up **1** High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) **1** Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) **1** Birds of prey are at the top of the food chain so receive the highest concentrations **1** Sub-lethal doses caused thin eggshells that broke / reduced breeding success – higher doses caused sterility or death **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **trophic level** → the feeding level in a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly **1 Disadvantage:** high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase) – chronic effects (memory loss, depression) – suspected carcinogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	2
04.3	1 mark for the point below. **1** Pyrethroids are highly toxic to fish (even though mammal toxicity is low) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) **1** They are moderately persistent so remain in the soil long enough to be absorbed **1** Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats **1** So bees / pollinators are less exposed than when the crop is sprayed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
05.1	**1 mark each** 1 mark for each of the 5 points below. **1** Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions **1** Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected – recirculation increases the metal concentration **1** Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) **1** Smelting needs chemical reduction at high temperatures – energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic **1** Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **Leachate** → polluted liquid draining out of waste **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves **1** The vegetation is harvested and incinerated **1** The concentrated metal in the ash is dissolved in acid and separated by electrolysis **OR** decontaminates polluted sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) **1** The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
06.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Asbestos:** fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma – disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) **2 Cyanide:** very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested – disposal: incineration so carbon and nitrogen are separated and oxidised to CO2 and NOx (much smaller problem) **2 High-level radioactive waste:** intense ionising radiation, heat generation, long half-lives – disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **vitrification** → sealing waste into glass **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
06.2	1 mark for the point below. **1** The fibres are bound / cannot become airborne so cannot be inhaled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material **1** The solid cement immobilises the waste so it cannot leak / leach out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
06.4	1 mark for the point below. **1** Even if the glass shatters the waste stays encapsulated in the fragments – glass is more stable / resistant to heat and leaching over very long timescales **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	
07.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 7×10^{11} tonnes **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10% of 10 000 = 1000 billion tonnes **1** 1000 / 700 x 100 = 143% (accept 142.9) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) **1** Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO_2 dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis **1** Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons **1** When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.5	**1 mark each** 1 mark for each of the 2 points below. **1** Aerobic respiration uses oxygen to break down organic compounds completely to CO_2 (and water), releasing much more energy **1** Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH_4), which is later oxidised to CO_2 in the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest (60 t ha^{-1} yr^{-1}) - over 100 times more **1** Ploughing downslope loses nearly three times as much as contour ploughing for the same crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **contour ploughing** → ploughing along the slope rather than up and down it	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash **1** Roots bind soil particles together **1** Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles **1** Plants increase infiltration (root channels, organic matter) so less surface runoff **1** Organic matter / humus from vegetation binds particles into stable peds **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **windbreak** → a line of trees or hedge that slows the wind • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff **1** Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **contour ploughing** → ploughing along the slope rather than up and down it	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away **1** In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Disposal of items containing mercury:** batteries, thermometers, fluorescent lights **1** Chemical plants producing chlorine using mercury electrodes **1** Combustion of coal **1** Gold mining (amalgamation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
09.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) **1** Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters **1** Disposal / storage of wastes at high pH to reduce solubility **1** Encapsulation of mercury-containing hazardous waste in cement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are more soluble at low pH / less soluble at high pH **1** So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** The effects of two pollutants interact to produce a different (greater) effect than either alone **1** Cadmium and zinc – ozone and sulfur dioxide **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming ClO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • **Evidence of success:** ozone hole area and depth stabilised and beginning to shrink – CFC concentrations in the atmosphere falling – UV levels not rising further – global ozone loss limited to about 4% • **Reasons for success:** international recognition of serious consequences (skin cancer) – agreement of nearly every country – legally binding targets with a fund for developing countries – relatively few producers / products – alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap – no lifestyle change needed • **Limits:** persistence of CFCs means recovery takes decades – some ODSs still used – HFC replacements are greenhouse gases • **Climate change contrast:** greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country – fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations – emissions still rising – natural fluctuations, time delays and tipping points make cause-effect less clear to the public	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Judgement:** ozone control was a technical substitution problem – climate change is a systemic one, so the ozone success is only partly transferable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • The converse of any point, ie the same comparison written the other way round. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **persistence** → does not break down / stays in the environment for a long time • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	
	Model answer - Level 3 (9 marks)	
	Ozone depletion followed the scientific method almost textbook-fashion: Antarctic UV and ozone measurements identified a problem, Rowland and Molina's 1974 hypothesis explained it, Dobson-unit, satellite and balloon data confirmed chlorine monoxide in the stratosphere, and the Montreal Protocol enacted a solution. It has largely worked: CFC concentrations are falling, the ozone hole has stabilised and is starting to shrink, and global ozone loss was limited to about 4%.	
	Success came from particular conditions. The consequence - skin cancer - was clear and frightening, nearly every country agreed, targets were legally binding with a fund to help developing countries, there were relatively few producers and products, and alternatives (HCFCs, HFCs, hydrocarbons, pump sprays) already existed, so substitution cost little. Even so recovery will take decades because CFCs are so persistent, some ozone-depleting substances are still in use, and the HFC replacements are themselves greenhouse gases.	
	Climate change has none of these advantages. Greenhouse gases come from almost every activity - energy, transport, agriculture, industry - in every country, and fossil fuels underpin entire economies, so control means changing lifestyles and economies rather than swapping one chemical for another. Kyoto and Paris set weaker, largely non-binding targets, developed and developing nations dispute responsibility, and natural fluctuations, time lags and tipping points make cause and effect less obvious to the public. Emissions are still rising. The ozone story shows international agreement can work, but it was a technical substitution problem; climate change is systemic, so its lessons transfer only partly.	
	Weaker answer - Level 2 (4 marks)	
	Ozone depletion was caused by CFCs. Scientists found the hole over Antarctica and the Montreal Protocol banned CFCs, so the ozone layer is now recovering. This was successful because everyone agreed and there were alternatives to CFCs. Climate change has not been tackled as well. There have been agreements like Kyoto and Paris but carbon dioxide is still going up. This is because fossil fuels are used everywhere and countries do not agree. So ozone depletion was solved better than climate change.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• Level 2 gives the right outline but Level 3 supports it with evidence (falling CFC levels, 4% loss, Dobson units, ClO detection) and explains each reason for the difference.	
	• Level 3 acknowledges the limits of the ozone success (persistence, HFCs) - balance is expected at the top level.	
	• The final judgement in Level 3 characterises the difference (substitution vs systemic problem); Level 2 just restates the question.	
10.2	1 mark for the point below. **1** Vienna Convention for the Protection of the Ozone Layer - no legally binding goals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ozone Layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

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